solutions

September 26, 2026

1 Chapter 3 Linear Regression

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1, 3, 10, 13

1.1 Conceptual

1. Describe the null hypotheses to which the p-values given in Table 3.4 correspond. Explain what conclusions you can draw based on these p-values. Your explanation should be phrased in terms of sales, TV, radio, and newspaper, rather than in terms of the coefficients of the linear model.

Table 3.4 is as follows:

	Coefficient	Std. Error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	<0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

3. Suppose we have a data set with five predictors, X_1 = GPA, X_2 = IQ, X_3 = Level (1 for College and 0 for High School), X_4 = Interaction between GPA and IQ, and X_5 = Interaction between GPA and Level. The response is starting salary after graduation (in thousands of dollars). Suppose we use least squares to fit the model, and get

$$\hat{\beta}_0 = 50, \hat{\beta}_1 = 20, \hat{\beta}_2 = 0.07, \hat{\beta}_3 = 35, \hat{\beta}_4 = 0.01, \hat{\beta}_5 = -10$$

   a. Which answer is correct, and why?

   b. For a fixed value of IQ and GPA, high school graduates earn more, on averae, than college graduates.

   ii. For a fixed value of IQ and GPA, college graduates earn more, on average, than high school graduates.

   iii. For a fixed value of IQ and GPA, high school graduates earn more, on average, than college graduates provided that the GPA is high enough.

iv. For a fixed value of IQ and GPA, college graduates earn more, on average, than high school graduates provided that the GPA is high enough.

b. Predict the salary of a college graduate with IQ of 110 and a GPA of 4.0.

c. True or false: Since the coefficient for the GPA/IQ interaction term is very small, there is very little evidence of an interaction effect. Justify your answer.

1.2 Applied

10. This question should be answered using the Carseats data set.

    (a) Fit a multiple regression model to predict `Sales` using `Price`, `Urban`, and `US`.

    (b) Provide an interpretation of each coefficient in the model. Be careful—some of the variables in the model are qualitative!

    (c) Write out the model in equation form, being careful to handle the qualitative variables properly.

    (d) For which of the predictors can you reject the null hypothesis $H_0 : \beta_j = 0$?

    (e) On the basis of your response to the previous question, fit a smaller model that only uses the predictors for which there is evidence of association with the outcome.

    (f) How well do the models in (a) and (e) fit the data?

    (g) Using the model from (e), obtain 95% confidence intervals for the coefficient(s).

    (h) Is there evidence of outliers or high leverage observations in the model from (e)?

11. (!!QUESTION THIRTEEN!!) In this exercise you will create some simulated data and will fit simple linear regression models to it. Make sure to use the default random number generator with seed set to 1 prior to starting part (a) to ensure consistent results.

    (a) Using the `normal()` method of your random number generator, create a vector, `x`, containing 100 observations drawn from a N(0, 1) distribution. This represents a feature, X.

    (b) Using the `normal()` method, create a vector, `eps`, containing 100 observations drawn from a N(0, 0.25) distribution—a normal distribution with mean zero and variance 0.25.

    (c) Using `x` and `eps`, generate a vector `y` according to the model (3.39)

$$Y = -1 + 0.5X + \epsilon$$

    What is the length of the vector `y`? What are the values of β_0 and β_1 in this linear model?

    (d) Create a scatterplot displaying the relationship between `x` and `y`. Comment on what you observe.

    (e) Fit a least squares linear model to predict `y` using `x`. Comment on the model obtained. How do $\widehat{\beta}_0$ and $\widehat{\beta}_1$ compare to β_0 and β_1?

(f) Display the least squares line on the scatterplot obtained in (d). Draw the population regression line on the plot, in a different color. Use the `legend()` method of the axes to create an appropriate legend.

(g) Now fit a polynomial regression model that predicts `y` using `x` and `x^2`. Is there evidence that the quadratic term improves the model fit? Explain your answer.

(h) Repeat (a)–(f) after modifying the data generation process in such a way that there is *less* noise in the data. The model (3.39) should remain the same. You can do this by decreasing the variance of the normal distribution used to generate the error term ϵ in (b). Describe your results.

(i) Repeat (a)–(f) after modifying the data generation process in such a way that there is *more* noise in the data. The model (3.39) should remain the same. You can do this by increasing the variance of the normal distribution used to generate the error term ϵ in (b). Describe your results.

(j) What are the confidence intervals for β_0 and β_1 based on the original data set, the noisier data set, and the less noisy data set? Comment on your results.

1.3 Answers

1. Each p-value tests the null hypothesis that the corresponding predictor has no linear association with sales, holding the other predictors constant.

   - Intercept: The null hypothesis is that expected sales are zero when TV, radio, and newspaper advertising spending are all zero. Since ($p < 0.0001$), we reject this hypothesis.
   - TV: The null hypothesis is that TV advertising spending is not associated with sales after accounting for radio and newspaper spending. Since ($p < 0.0001$), there is strong evidence that TV advertising is associated with sales. We reject this too.
   - Radio: The null hypothesis is that radio advertising spending is not associated with sales after accounting for TV and newspaper spending. Since ($p < 0.0001$), there is strong evidence that radio advertising is associated with sales. Once again, we reject.
   - Newspaper: The null hypothesis is that newspaper advertising spending is not associated with sales after accounting for TV and radio spending. Since ($p = 0.8599$), we fail to reject the null hypothesis. There is no significant evidence that newspaper advertising is associated with sales in this model. We don't reject.

2. ***For question 3*** (markdown formatting isn't properly numbering this list for some reason, so ignore the number on the left)

   a. The correct answer is **iii**.

   For fixed IQ and GPA, the difference between college and high school graduates is:

   $$\text{College} - \text{High School} = \beta_3 + \beta_5(\text{GPA}) = 35 - 10(\text{GPA})$$

   College graduates earn more when $35 - 10\text{GPA} > 0$, which means $\text{GPA} < 3.5$. High school graduates earn more when $35 - 10\text{GPA} < 0$, which means $\text{GPA} > 3.5$.

   So high school graduates earn more, on average, than college graduates provided the GPA is high enough. That matches part **iii**.

b. For a college graduate, Level = 1, GPA = 4.0, and IQ = 110.

$$\begin{aligned}\widehat{\text{Salary}} &= 50 + 20(4.0) + 0.07(110) + 35(1) + 0.01(4.0)(110) - 10(4.0)(1) \\ &= 50 + 80 + 7.7 + 35 + 4.4 - 40 \\ &= 137.1\end{aligned}$$

So the predicted starting salary is \$137,100.

c. False because a small coefficient does not by itself mean there is little evidence of an interaction effect. The evidence depends on the size of the coefficient relative to its stamdard error. A small coefficient like 0.01 can still be statistically significant if its standard error is small enough. Since no standard error is given here, we cannot say that there is little evidence of an interaction effect just because the coefficient is small.

[2]:
```
# 10
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf
from ISLP import load_data

Carseats = load_data("Carseats")

# (a) Full model
full_model = smf.ols("Sales ~ Price + C(Urban) + C(US)", data=Carseats).fit()
print(full_model.summary())

# (d) Coefficient tests
print("\nP-values:")
print(full_model.pvalues)

# (e) Reduced model: retain Price and US
reduced_model = smf.ols("Sales ~ Price + C(US)", data=Carseats).fit()
print("\nReduced model:")
print(reduced_model.summary())

# (f) Compare model fit
print("\nModel comparison:")
print(pd.DataFrame({
    "Model": ["Full", "Reduced"],
    "R-squared": [full_model.rsquared, reduced_model.rsquared],
    "Adjusted R-squared": [
        full_model.rsquared_adj,
        reduced_model.rsquared_adj
    ],
    "Residual standard error": [
        np.sqrt(full_model.mse_resid),
```

```
        np.sqrt(reduced_model.mse_resid)
    ]
}))

# (g) 95% confidence intervals for the reduced model
print("\n95% confidence intervals:")
print(reduced_model.conf_int(alpha=0.05))

# (h) Outlier and leverage diagnostics
influence = reduced_model.get_influence()
summary_frame = influence.summary_frame()

n = len(Carseats)
p = int(reduced_model.df_model) + 1
leverage_cutoff = 2 * p / n

diagnostics = pd.DataFrame({
    "studentized_residual": summary_frame["student_resid"],
    "leverage": summary_frame["hat_diag"],
    "cooks_distance": summary_frame["cooks_d"]
})

print("\nPotential outliers:")
print(diagnostics[diagnostics["studentized_residual"].abs() > 2])

print("\nPotential high-leverage observations:")
print(diagnostics[diagnostics["leverage"] > leverage_cutoff])

# Diagnostic plots
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].scatter(reduced_model.fittedvalues, reduced_model.resid)
axes[0].axhline(0, color="red", linestyle="--")
axes[0].set_xlabel("Fitted values")
axes[0].set_ylabel("Residuals")
axes[0].set_title("Residuals vs. Fitted Values")

sm = influence.summary_frame()
axes[1].scatter(np.arange(n), sm["cooks_d"])
axes[1].set_xlabel("Observation")
axes[1].set_ylabel("Cook's distance")
axes[1].set_title("Cook's Distance")

plt.tight_layout()
plt.show()

# (h) The residual plot shows observations with studentized residuals
```

```
# greater than 2 in absolute value, which means there are possible outliers. ␣
 ↪However,
# the residuals are generally centered around zero and show no strong pattern.
# Many observations have relatively large Cook's distances compared with
# the 4/n guideline, but all are far below 1, so none appears highly
# influential.
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.239
Model:                            OLS   Adj. R-squared:                  0.234
Method:                 Least Squares   F-statistic:                     41.52
Date:                Sat, 26 Sep 2026   Prob (F-statistic):           2.39e-23
Time:                        17:23:56   Log-Likelihood:                -927.66
No. Observations:                 400   AIC:                             1863.
Df Residuals:                     396   BIC:                             1879.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
---
Intercept          13.0435      0.651     20.036      0.000      11.764
14.323
C(Urban)[T.Yes]    -0.0219      0.272     -0.081      0.936      -0.556
0.512
C(US)[T.Yes]        1.2006      0.259      4.635      0.000       0.691
1.710
Price              -0.0545      0.005    -10.389      0.000      -0.065
-0.044
==============================================================================
Omnibus:                        0.676   Durbin-Watson:                   1.912
Prob(Omnibus):                  0.713   Jarque-Bera (JB):                0.758
Skew:                           0.093   Prob(JB):                        0.684
Kurtosis:                       2.897   Cond. No.                         628.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.

P-values:
Intercept          3.626602e-62
C(Urban)[T.Yes]    9.357389e-01
C(US)[T.Yes]       4.860245e-06
Price              1.609917e-22
```

```
dtype: float64

Reduced model:
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.239
Model:                            OLS   Adj. R-squared:                  0.235
Method:                 Least Squares   F-statistic:                     62.43
Date:                Sat, 26 Sep 2026   Prob (F-statistic):           2.66e-24
Time:                        17:23:56   Log-Likelihood:                -927.66
No. Observations:                 400   AIC:                             1861.
Df Residuals:                     397   BIC:                             1873.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      13.0308      0.631     20.652      0.000      11.790      14.271
C(US)[T.Yes]    1.1996      0.258      4.641      0.000       0.692       1.708
Price          -0.0545      0.005    -10.416      0.000      -0.065      -0.044
==============================================================================
Omnibus:                        0.666   Durbin-Watson:                   1.912
Prob(Omnibus):                  0.717   Jarque-Bera (JB):                0.749
Skew:                           0.092   Prob(JB):                        0.688
Kurtosis:                       2.895   Cond. No.                         607.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.

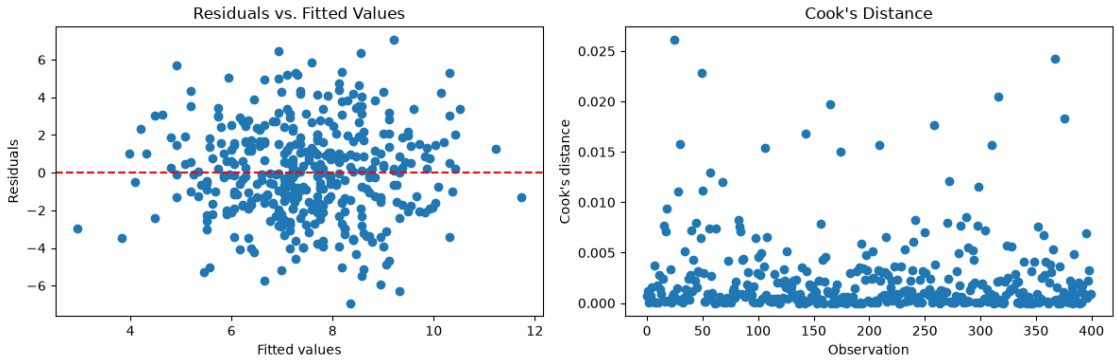
Model comparison:
     Model  R-squared  Adjusted R-squared  Residual standard error
0     Full   0.239275            0.233512                 2.472492
1  Reduced   0.239263            0.235430                 2.469397

95% confidence intervals:
                     0         1
Intercept     11.79032  14.271265
C(US)[T.Yes]   0.69152   1.707766
Price         -0.06476  -0.044195

Potential outliers:
     studentized_residual  leverage  cooks_distance
17               2.118504  0.004779        0.007122
25               2.599652  0.011622        0.026109
28              -2.433745  0.005636        0.011054
30               2.194942  0.009835        0.015799
```

```
49               2.334376  0.012553        0.022835
50              -2.835843  0.004224        0.011173
57              -2.348788  0.007084        0.012972
62              -2.105923  0.005052        0.007442
68               2.642364  0.005202        0.011988
82               2.023359  0.006085        0.008290
98               2.110650  0.004342        0.006420
106             -2.166285  0.009857        0.015429
143             -2.060961  0.011861        0.016857
209             -2.581586  0.007100        0.015663
258             -2.106350  0.011913        0.017677
297             -2.241569  0.004612        0.007683
298              2.054143  0.008198        0.011532
304              2.037942  0.005202        0.007182
316              2.179402  0.012883        0.020470
328             -2.050664  0.004027        0.005623
352              2.390178  0.003997        0.007552
376              2.891521  0.006637        0.018282
395              2.166210  0.004438        0.006908

Potential high-leverage observations:
     studentized_residual  leverage  cooks_distance
42              -0.534993  0.043338    4.329756e-03
125             -0.418660  0.025966    1.560768e-03
155             -0.637155  0.016106    2.218521e-03
156              1.230079  0.015356    7.855441e-03
159              0.041843  0.015707    9.336799e-06
165             -1.421459  0.028567    1.975504e-02
171              0.513511  0.021014    1.890242e-03
174             -1.214486  0.029687    1.502431e-02
191              0.761418  0.018039    3.553907e-03
203             -0.990770  0.015356    5.103071e-03
208             -0.720603  0.018235    3.218758e-03
269              0.417664  0.019195    1.140355e-03
272              1.383849  0.018687    1.212821e-02
313             -0.310632  0.023165    7.644786e-04
315              0.602083  0.017049    2.099193e-03
356             -0.210829  0.018279    2.765330e-04
365              0.949537  0.017399    5.322958e-03
367              1.736609  0.023707    2.428736e-02
383              0.009660  0.016514    5.236050e-07
386              0.410219  0.016555    9.462170e-04
```



[3]:
```python
# 13
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm

# Use the default random number generator with seed 1.
def generate_data(error_sd):
    rng = np.random.default_rng(1)
    x = rng.normal(loc=0, scale=1, size=100)
    eps = rng.normal(loc=0, scale=error_sd, size=100)
    y = -1 + 0.5 * x + eps
    return x, eps, y

# (a)-(c): original data, where Var(eps) = 0.25.
x, eps, y = generate_data(error_sd=0.5)
print(f"(c) Length of y: {len(y)}")
print("    beta_0 = -1 and beta_1 = 0.5")

# (d) Scatterplot.
fig, ax = plt.subplots(figsize=(6, 4))
ax.scatter(x, y, alpha=0.75)
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_title("Original data: y versus x")
plt.show()

# (e) Fit the least-squares model.
X = sm.add_constant(x)
original_model = sm.OLS(y, X).fit()
print("\n(e) Original least-squares estimates:")
print(original_model.summary())
print(f"Estimated beta_0 = {original_model.params[0]:.4f}")
```

```
print(f"Estimated beta_1 = {original_model.params[1]:.4f}")

# (f) Least-squares line and population regression line.
x_grid = np.linspace(x.min(), x.max(), 200)
fig, ax = plt.subplots(figsize=(6, 4))
ax.scatter(x, y, alpha=0.6, label="Observed data")
ax.plot(x_grid, original_model.params[0] + original_model.params[1] * x_grid,
        color="red", label="Least-squares line")
ax.plot(x_grid, -1 + 0.5 * x_grid,
        color="black", linestyle="--", label="Population regression line")
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_title("Least-squares and population regression lines")
ax.legend()
plt.show()

# (g) Polynomial regression with x and x^2.
polynomial_model = sm.OLS(y, sm.add_constant(np.column_stack((x, x**2)))).fit()
print("\n(g) Polynomial model:")
print(polynomial_model.summary())
print(f"p-value for x^2 = {polynomial_model.pvalues[2]:.4f}")

# (h)-(i): repeat the simulation with less and more noise.
def fit_simulation(error_sd):
    x_sim, eps_sim, y_sim = generate_data(error_sd)
    model_sim = sm.OLS(y_sim, sm.add_constant(x_sim)).fit()
    return x_sim, y_sim, model_sim

less_x, less_y, less_model = fit_simulation(error_sd=0.1)
more_x, more_y, more_model = fit_simulation(error_sd=1.0)

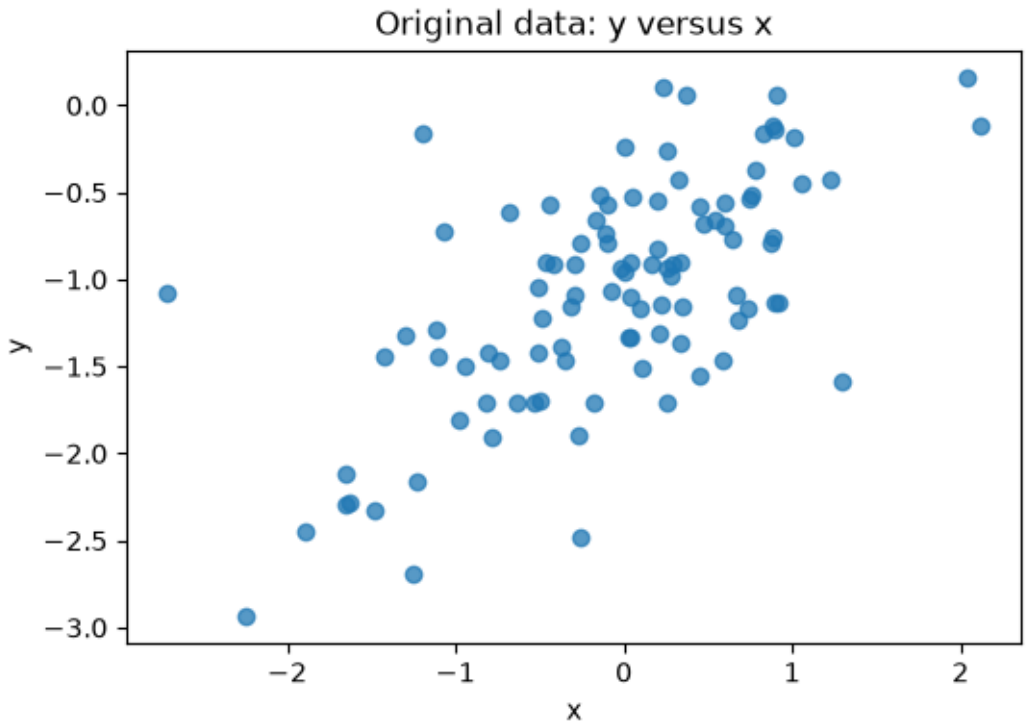
print("\n(h)-(i) Comparison of noise levels:")
comparison = pd.DataFrame({
    "Data set": ["Less noise", "Original", "More noise"],
    "Error variance": [0.01, 0.25, 1.00],
    "Estimated intercept": [less_model.params[0], original_model.params[0],␣
 ↪more_model.params[0]],
    "Estimated slope": [less_model.params[1], original_model.params[1],␣
 ↪more_model.params[1]],
    "R-squared": [less_model.rsquared, original_model.rsquared, more_model.
 ↪rsquared],
    "Residual SD": [np.sqrt(less_model.mse_resid), np.sqrt(original_model.
 ↪mse_resid), np.sqrt(more_model.mse_resid)]
})
print(comparison.round(4).to_string(index=False))

fig, axes = plt.subplots(1, 3, figsize=(15, 4), sharey=True)
```

```
for ax, x_sim, y_sim, model_sim, title in [
    (axes[0], less_x, less_y, less_model, "Less noise"),
    (axes[1], x, y, original_model, "Original noise"),
    (axes[2], more_x, more_y, more_model, "More noise")
]:
    ax.scatter(x_sim, y_sim, alpha=0.65)
    grid = np.linspace(x_sim.min(), x_sim.max(), 200)
    ax.plot(grid, model_sim.params[0] + model_sim.params[1] * grid, color="red")
    ax.plot(grid, -1 + 0.5 * grid, color="black", linestyle="--")
    ax.set_title(title)
    ax.set_xlabel("x")
axes[0].set_ylabel("y")
plt.tight_layout()
plt.show()

# (j) Confidence intervals for beta_0 and beta_1.
ci = pd.DataFrame({
    "Data set": ["Less noise", "Original", "More noise"],
    "beta_0 lower": [less_model.conf_int()[0, 0], original_model.conf_int()[0,␣
 ↪0], more_model.conf_int()[0, 0]],
    "beta_0 upper": [less_model.conf_int()[0, 1], original_model.conf_int()[0,␣
 ↪1], more_model.conf_int()[0, 1]],
    "beta_1 lower": [less_model.conf_int()[1, 0], original_model.conf_int()[1,␣
 ↪0], more_model.conf_int()[1, 0]],
    "beta_1 upper": [less_model.conf_int()[1, 1], original_model.conf_int()[1,␣
 ↪1], more_model.conf_int()[1, 1]]
})
print("\n(j) 95% confidence intervals:")
print(ci.round(4).to_string(index=False))
```

```
(c) Length of y: 100
    beta_0 = -1 and beta_1 = 0.5
```



(e) Original least-squares estimates:

OLS Regression Results

Dep. Variable:	y	R-squared:	0.409
Model:	OLS	Adj. R-squared:	0.403
Method:	Least Squares	F-statistic:	67.79
Date:	Sat, 26 Sep 2026	Prob (F-statistic):	8.04e-13
Time:	18:05:04	Log-Likelihood:	-71.745
No. Observations:	100	AIC:	147.5
Df Residuals:	98	BIC:	152.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-1.0380	0.050	-20.647	0.000	-1.138	-0.938
x1	0.4843	0.059	8.233	0.000	0.368	0.601

Omnibus:	1.277	Durbin-Watson:	2.198
Prob(Omnibus):	0.528	Jarque-Bera (JB):	0.759
Skew:	0.114	Prob(JB):	0.684
Kurtosis:	3.361	Cond. No.	1.20

```
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
Estimated beta_0 = -1.0380
Estimated beta_1 = 0.4843
```



(g) Polynomial model:

OLS Regression Results

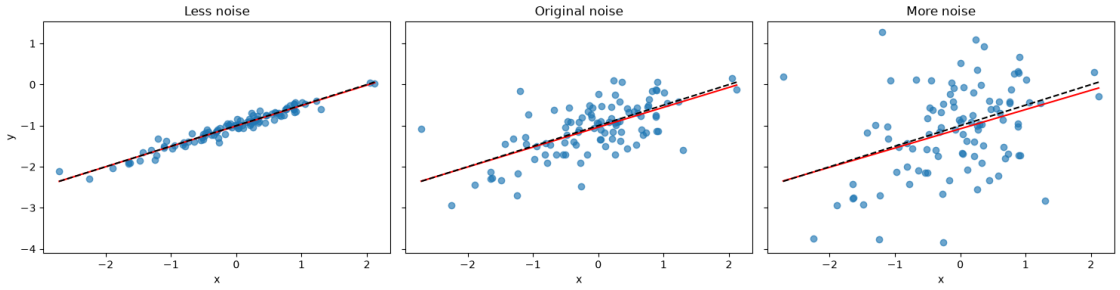
Dep. Variable:	y	R-squared:	0.409
Model:	OLS	Adj. R-squared:	0.397
Method:	Least Squares	F-statistic:	33.55
Date:	Sat, 26 Sep 2026	Prob (F-statistic):	8.42e-12
Time:	18:05:04	Log-Likelihood:	-71.744
No. Observations:	100	AIC:	149.5
Df Residuals:	97	BIC:	157.3
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]

```
const          -1.0364      0.060    -17.399      0.000      -1.155      -0.918
x1              0.4831      0.063      7.647      0.000       0.358       0.609
x2             -0.0024      0.045     -0.052      0.959      -0.092       0.088
==============================================================================
Omnibus:                        1.338   Durbin-Watson:                   2.197
Prob(Omnibus):                  0.512   Jarque-Bera (JB):                0.814
Skew:                           0.119   Prob(JB):                        0.666
Kurtosis:                       3.372   Cond. No.                         2.23
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
p-value for x^2 = 0.9588

(h)-(i) Comparison of noise levels:
  Data set  Error variance  Estimated intercept  Estimated slope  R-squared
Residual SD
Less noise            0.01              -1.0076           0.4969     0.9479
0.1002
  Original            0.25              -1.0380           0.4843     0.4089
0.5009
More noise            1.00              -1.0760           0.4686     0.1393
1.0018
```




```
(j) 95% confidence intervals:
  Data set  beta_0 lower  beta_0 upper  beta_1 lower  beta_1 upper
Less noise       -1.0276       -0.9876        0.4735        0.5202
  Original       -1.1378       -0.9382        0.3676        0.6010
More noise       -1.2756       -0.8765        0.2351        0.7020
```

1.4 Question 13: Written answers

- **(d)** The scatterplot shows a positive, approximately linear relationship between `x` and `y`, with noticeable random variation around the trend.

- **(e)** The estimated model is approximately

$$\hat{y} = -1.038 + 0.484x.$$

  These estimates are close to the population values -1 and 0.5, as expected from random sampling.

- **(g)** The estimated coefficient of x^2 is approximately -0.0024, with p-value 0.9588. Since this p-value is very large, there is no evidence that adding a quadratic term improves the model.

- **(h)** With less noise, the points lie much closer to the population line, the fitted coefficients are closer to -1 and 0.5, and R^2 is much larger.

- **(i)** With more noise, the points are more dispersed, the fitted line is less precise, and R^2 is smaller. The estimated coefficients can also deviate farther from their population values.

- **(j)** The 95% confidence intervals are:

Data set	95% CI for β_0	95% CI for β_1
Less noise	$(-1.0276, -0.9876)$	$(0.4735, 0.5202)$
Original noise	$(-1.1378, -0.9382)$	$(0.3676, 0.6010)$
More noise	$(-1.2756, -0.8765)$	$(0.2351, 0.7020)$

  The intervals become narrower when there is less noise and wider when there is more noise. All three intervals contain the true values $\beta_0 = -1$ and $\beta_1 = 0.5$.